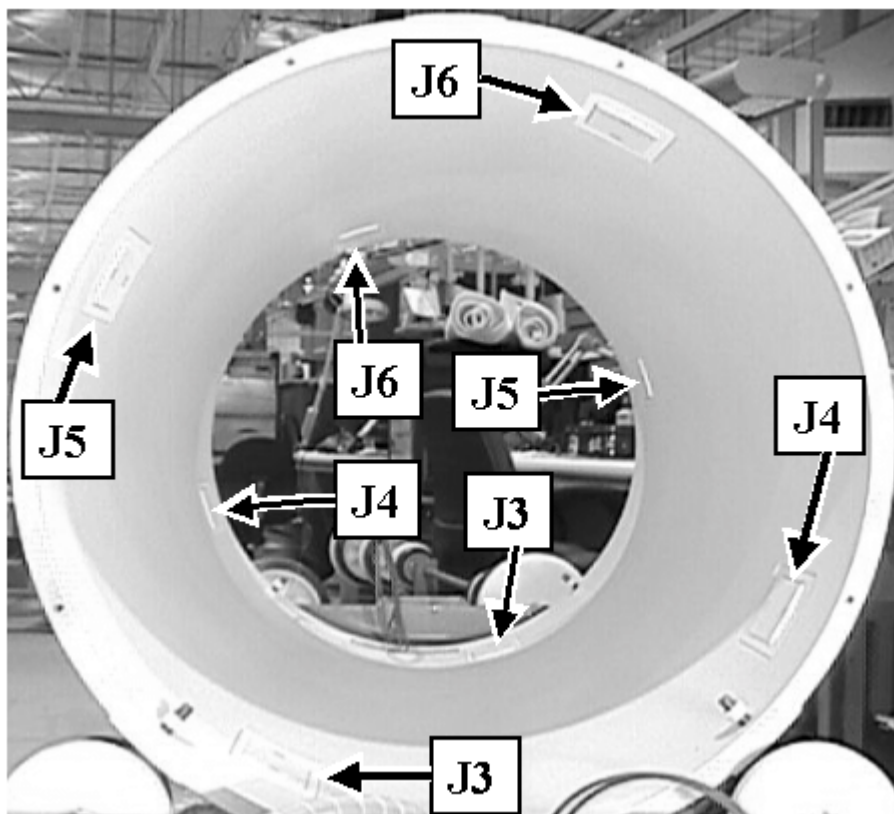


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1 - OVERVIEW

This procedure describes the process for replacing a failed dynamic disable diode board inside the 1.5T Twin RF body coil.



REAR VIEW INTO TWIN RF COIL SHOWING BOARD POSITIONS CORRESPONDING TO BIAS CABLES
ILLUSTRATION 1-1

Two cables protrude from each end of the RF coil. Each cable provides dynamic disable bias to two (2) diode boards on that end of the coil. The blue, rigid RF drive coax cables exit from the rear of the RF coil. Refer to Table 1-1 to see the biasing scheme for the dynamic disable diode boards.

TABLE 1-1
DYNAMIC DISABLE BIAS CONNECTIONS

COIL SIDE	BIAS CABLES	DIODE BOARD POSITION
FRONT	J3/J5	J3, J5 (FRONT)
FRONT	J4/J6	J4, J6 (FRONT)
REAR	J3/J5	J3, J5 (REAR)
REAR	J4/J6	J4, J6 (REAR)

2 - TOOLS REQUIRED

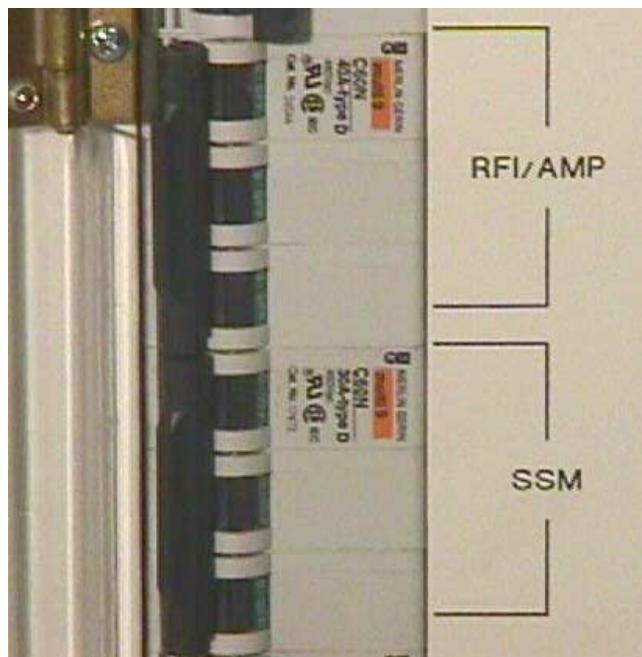
- 5/16" non-magnetic slot screwdriver (GE part # 46-320473P13)

3 – SRF CABINET POWER LOCKOUT AND TAGOUT



POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE SRF CABINET BEFORE ATTEMPTING TO SERVICE THE DYNAMIC DISABLE CIRCUIT. VERIFY THAT THE RFI/AMP AND SSM BREAKERS ON THE FRONT OF THE PDU ARE OFF. LOCK AND TAG OUT THE PLEXI-GLASS COVER ON THE FRONT OF THE PDU.

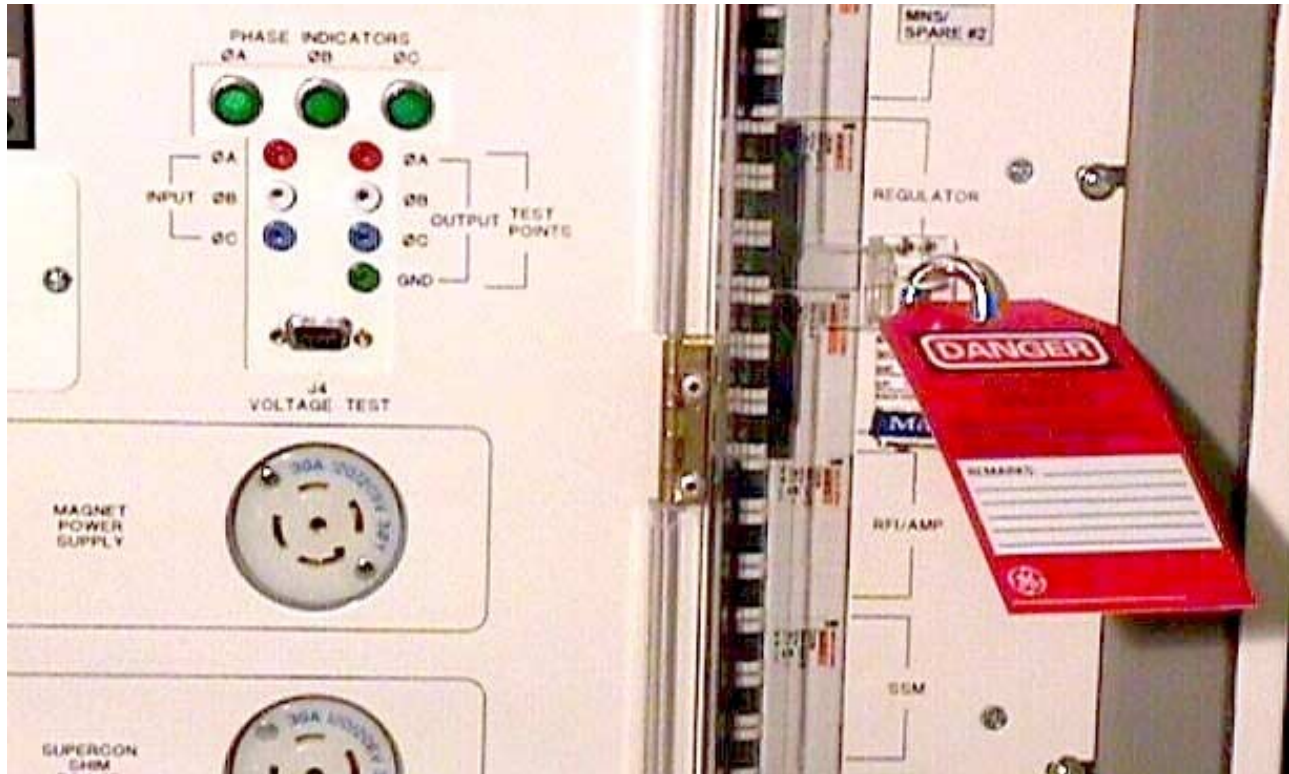
1. Remove the cover from the front of the ACGD Cabinet.
2. Open Plexiglas door over the front of the PDU breakers and switch off the **RFI/AMP** and **SSM** breakers. See Illustration 3-1.



BREAKER IDENTIFICATION
ILLUSTRATION 3-1

3 – SRF CABINET POWER LOCKOUT AND TAGOUT (CONTINUED)

3. Close the plexi-glass door and lock and tag it out. See Illustration 3-2. (Refer to CD-ROM *Dir. 2187583-3 [or -2], MR Release Signa 5x/8x Service Methods, Renewal Parts and Service Tools*, Safety, Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)

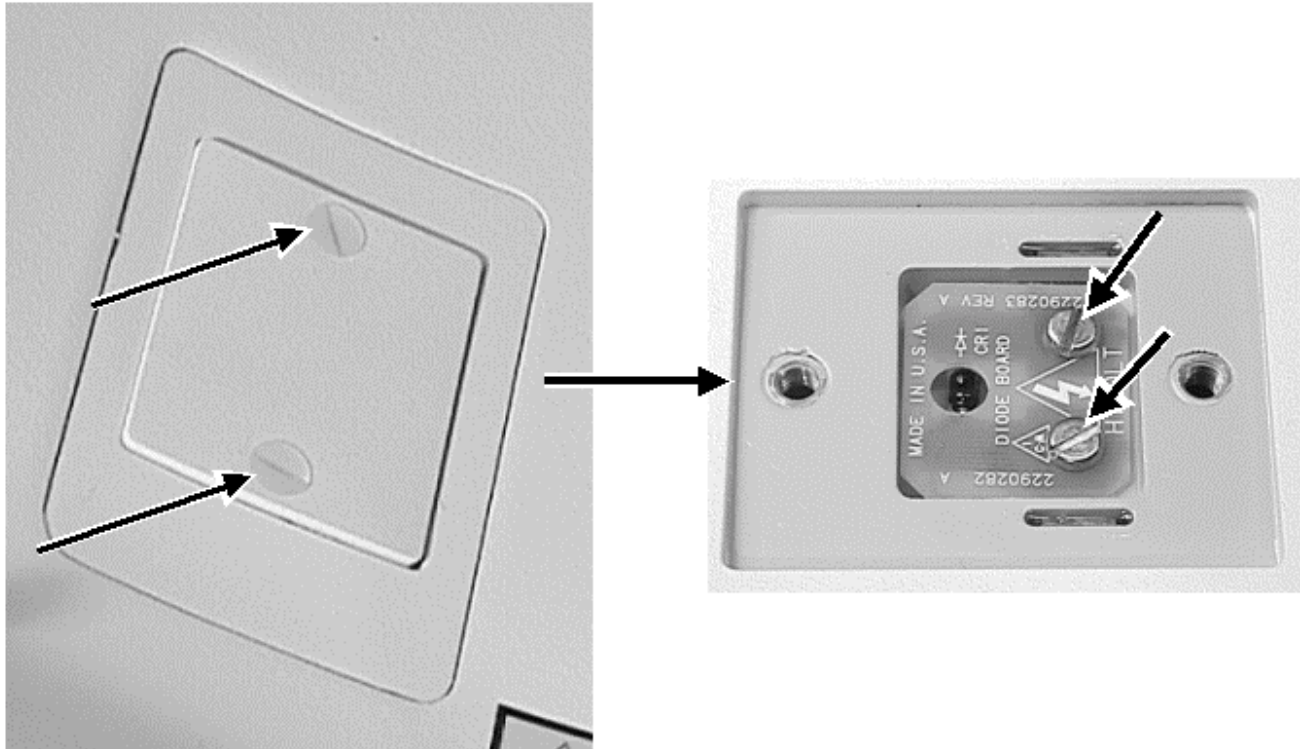


LOCKOUT AND TAGOUT OF BREAKERS
ILLUSTRATION 3-2

4. Remove the front cover from the SRF Cabinet.
5. Verify all LEDs are OFF (not illuminated) on front of SSM and RFI. Verify all LEDs are OFF (not illuminated) at the rear of the RF Amplifier Power Module.
6. Using an AC voltmeter (DVM), measure the input power leads on the rear of one of the RF Amplifier Modules from the rear (right side) of the RF/PDU Cabinet.
 - Measure between L1 and L2; verify voltmeter measures 0 VAC.

4 – DYNAMIC DISABLE DIODE BOARD REPLACEMENT

1. Perform **Section 3 – SRF CABINET POWER LOCKOUT AND TAGOUT** (Refer to *MR Signa OpenSpeed Service Methods CDRom*, Safety (SYSMSA2.DOC), Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.).
2. Using a 5/16" non-magnetic slot screwdriver, remove the two (2) plastic screws from the dynamic disable switch board cover plate and remove the plate to expose the diode board as shown in Illustration 4-1.



PROCESS FOR REMOVING DIODE BOARD FROM RF COIL
ILLUSTRATION 4-1

NOTE

The two (2) M4 X 8 brass screws securing the dynamic disable diode board to the coil are not held to the board with brass keepers. Keepers were not used to avoid a potential source of spike noise. Exercise care when removing the brass screws so as not to lose them under the bridge.

3. Refer to Illustration 4-1. Remove the two (2) brass screws from the diode board and remove the diode board from the coil.

NOTE

An extra brass screw has been included with the new diode board in case one of the old ones is somehow lost. If not needed or used, this extra screw can be discarded.

4 – DYNAMIC DISABLE DIODE BOARD REPLACEMENT (CONTINUED)

4. Locate the 0.5 ml capsule of Loctite™ 242 that was shipped with the diode board and apply one drop of the blue fluid to the threads of each brass screw.
5. Install the new diode board in the reverse order from how it was removed.



The purpose of the diode board brass screws is to securely hold the diode board against the foil traces on the RF coil. The screws were not meant to be a part of the electrical circuit. The diode board must always be installed with the foil traces against the traces on the RF coil to make good electrical contact.

- a. Install the diode board in the RF coil with the foil side down against the traces on the RF coil.
- b. Secure the diode board to the RF coil with the brass screws.



The plastic screws that secure the cover over the dynamic disable diode boards can be easily broken if over-tightened.

- c. Secure the cover over the dynamic disable diode board using the plastic screws.

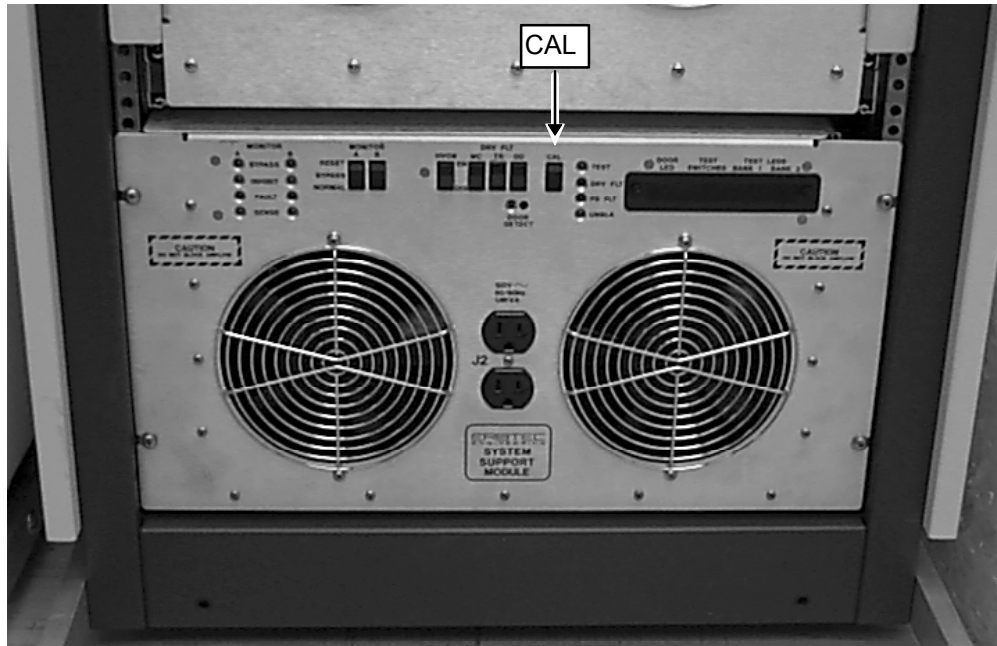
5 – SYSTEM RESTORATION

1. Remove lock and tag from the plexi-glass cover on the front of the PDU cabinet and put the RFI/AMP and the SSM breakers into the right (ON) position.
2. Confirm that LEDs are illuminated on the front of the RFI and SSM and confirm that the fans in the RF amplifiers are rotating.

WARNING!

THE SSM WILL NOT FAULT IF IT IS CALIBRATED WITH A DISCONNECTION OR POOR CONNECTION TO ONE OF THE 4 MAGNET ROOM DYNAMIC DISABLE BIAS LINES (J72, J73, J75, AND J76 ON THE MAGNET ROOM SIDE OF THE PENETRATION PANEL). IMAGE SNR DEGRADATION WILL BE NOTICED BUT THE SSM MAY NOT FAULT. MAKE CERTAIN THE FOUR BIAS LINE CONNECTIONS FROM THE PEN PANEL TO THE RF COIL ARE CONNECTED BEFORE PERFORMING THIS CALIBRATION.

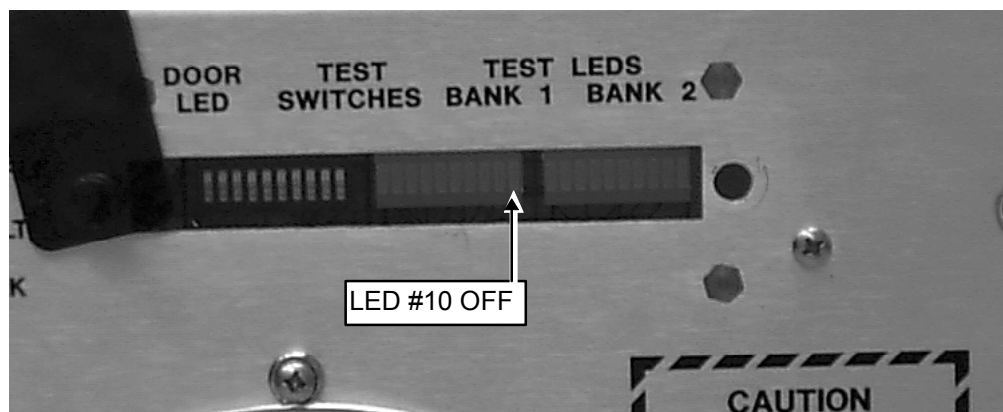
3. Press and release the CAL switch on the front of the SSM (see Illustration 5-1). This samples the open circuit fault detect signals and adjusts the threshold accordingly for the Dynamic Disable Circuits: J42 (Body 1), J43 (Body 2), and J44 (Direct Drive). When the CAL switch is pressed the LEDs labeled TEST, DRV FLT, and PS FLT will illuminate (this does not indicate a fault).



SSM CAL SWITCH LOCATION
ILLUSTRATION 5-1

5 – SYSTEM RESTORATION (CONTINUED)

4. View the front panel LEDs. Ensure the Bank 1, LED #10 is OFF (see Illustration 5-2). This LED indicates the adjustment process was successful.



SSM FRONT PANEL LEDs
ILLUSTRATION 5-2

5. Perform one head and one body scan to verify that the system is functional.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	August 10, 2001	Don Thome'	Initial version.